

INTEGRATION of e^x and $e^{(ax+b)}$ → must be 'linear'

We know $\frac{d}{dx}(e^x) = e^x$

so, $\int e^x dx = e^x + c$

what about $\int e^{2x} dx$

$$\int e^{2x} dx = \frac{1}{2} e^{2x} + c \quad \left[\begin{array}{l} \frac{d}{dx}(e^{2x}) \\ = e^{2x} \cdot (2) \end{array} \right]$$

so, in general $\int e^{(ax+b)} dx = \frac{1}{a} e^{(ax+b)} + c$

Example:

1. $\int e^{3x} dx = \frac{1}{3} e^{3x} + c$

2. $\int e^{1-2x} dx = -\frac{1}{2} e^{1-2x} + c$

$\int \square e^{\square} dx$

3. $\int \frac{1}{4e^{3x}} dx = \int \frac{1}{4} e^{-3x} dx = \frac{1}{4} \times e^{-3x} \times \frac{1}{-3} + c$
 $= -\frac{1}{12} e^{-3x} + c = -\frac{1}{12e^{3x}} + c$

4. $\int_1^3 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_1^3 = \left(\frac{1}{2} e^6 \right) - \left(\frac{1}{2} e^2 \right) = \frac{1}{2} (e^6 - e^2)$

INTEGRATION OF $\frac{1}{x}$ and $\frac{1}{ax+b}$

we know $\frac{d}{dx} (\ln x) = \frac{1}{x}$

so, $\int \frac{1}{x} dx = \ln|x| + c$



$$\int x^{-1} dx = \ln|x| + c$$

it doesn't work on
 $\int x^{-1} dx = \frac{x^{-1+1}}{-1+1}$

and $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + c$

Example:

1. $\int \frac{1}{x+3} dx = \frac{1}{1} \ln|x+3| + c$

2. $\int \frac{1}{2x+3} dx = \frac{1}{2} \ln|2x+3| + c$

3. $\int \frac{2}{3x-5} dx = \frac{2}{3} \ln|3x-5| + c$

4. $\int_0^2 \frac{1}{5x+3} dx = \left[\frac{1}{5} \ln|5x+3| \right]_0^2$
 $= \left[\frac{1}{5} \ln|5(2)+3| \right] - \left[\frac{1}{5} \ln|5(0)+3| \right]$
 $= \left[\frac{1}{5} \ln|13| \right] - \left[\frac{1}{5} \ln|3| \right]$
 $= \frac{1}{5} \ln 13 - \frac{1}{5} \ln(3) = \frac{1}{5} (\ln 13 - \ln 3)$
 $= \frac{1}{5} \ln\left(\frac{13}{3}\right)$ Exact value
 $= 0.293$ (3sf)

$$\int_{-1}^3 \frac{4x^2+8x}{2x+3} dx$$

↳ to integrate fractions like this, first, use Long Division

$$\frac{4x^2+8x}{2x+3} = 2x+1 + \frac{-3}{2x+3}$$

$$\int_{-1}^3 2x+1 - \frac{3}{2x+3} dx$$

INTEGRATION OF Trig TERMS

$f(x)$

$\sin x$

$\cos x$

$\sec^2 x$

$\int f(x) dx$

$-\cos x$

$\sin x$

$\tan x$

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx$$

$$= -1 \int \frac{f(x)}{f(x)} dx = -\ln|\cos x| + c$$

$$= \ln(\cos^{-1} x) - \ln\left(\frac{1}{\cos x}\right) + c$$

$$= \ln(\sec x) + c$$

If the variable contains a linear term $(ax + b)$ the result are:

$$\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + c$$

Not on

formula sheet

$$\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + c$$

$$\int \sec^2(ax + b) dx = \frac{1}{a} \tan(ax + b) + c$$

Examples:

1. $\int \sin(2x) dx = -\frac{1}{2} \cos(2x) + c$

← Linear

check: $-\frac{1}{2}(-\sin 2x)(2) = \sin 2x$

2. $\int 2 \sec^2(5x - 2) dx = 2 \tan(5x - 2) \left(\frac{1}{5}\right) + c$

$$= \frac{2}{5} \tan(5x - 2) + c$$

Always work
in RADIANS!

3. $\int_0^{\frac{\pi}{3}} \sin 2x - \cos 4x dx = \left[-\frac{1}{2} \cos 2x - \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{3}}$

$$= \left[-\frac{1}{2} \cos \frac{2\pi}{3} - \frac{1}{4} \sin \frac{4\pi}{3} \right] - \left[-\frac{1}{2} \cos 0 - \frac{1}{4} \sin 0 \right]$$

$$= \left[-\frac{1}{2} \times -\frac{1}{2} - \frac{1}{4} \left(-\frac{\sqrt{3}}{2}\right) \right] - \left[-\frac{1}{2} (1) - \frac{1}{4} (0) \right]$$

$$= \frac{1}{4} + \frac{\sqrt{3}}{8} + \frac{1}{2} = \frac{6 + \sqrt{3}}{8}$$

4. $\int \sin(x^2) dx$ - not linear Can't do it.

INTEGRATION of $\sin^2 x$ or $\cos^2 x$ ($\sin x$)² or ($\cos x$)²
Not $\sin x^2$

To do this we use rearrangements of the double angle formulae to re-write $\sin^2 x$ or $\cos^2 x$ as expression with linear terms

Reminder of double angle formulae

$$\cos 2A \equiv \cos^2 A - \sin^2 A \equiv 2\cos^2 A - 1 \equiv 1 - 2\sin^2 A$$

$$\cos 2x = 2\cos^2 x - 1$$

$$\cos 2x + 1 = 2\cos^2 x$$

$$\frac{1}{2} \cos 2x + \frac{1}{2} = \cos^2 x$$

$$\frac{\cos 2x + 1}{2}$$

So $\int \cos^2 x \, dx = \int \frac{1}{2} \cos 2x + \frac{1}{2} \, dx$ linear

$$= \frac{1}{2} \times \frac{1}{2} \sin 2x + \frac{1}{2} x + C$$

$$= \frac{1}{4} \sin 2x + \frac{1}{2} x + C$$

$$\cos 2x = 1 - 2\sin^2 x$$

$$2\sin^2 x = 1 - \cos 2x$$

$$\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$$

so $\int \sin^2 x \, dx = \int \frac{1}{2} - \frac{1}{2} \cos 2x \, dx$

$$= \frac{1}{2} x - \frac{1}{2} \times \frac{1}{2} \sin 2x + C$$

$$= \frac{1}{2} x - \frac{1}{4} \sin 2x + C$$

INTEGRATION 'by PARTS'

from formula sheets :

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$u \frac{dv}{dx}$ is a product of terms and u is a term that can be differentiated and $\frac{dv}{dx}$ is a term that can be integrated.

Examples :

$$(\ln x)^2 \neq \ln(x^2)$$

$$1 \quad \int x \cos x \, dx$$

$\uparrow$ $\uparrow$
 u $\frac{dv}{dx}$

$$u = x$$
$$\frac{du}{dx} = 1$$

$$\frac{dv}{dx} = \cos x$$

$$v = \sin x$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int x \cos x \, dx = x \cdot \sin x - \int \sin x \cdot 1 \, dx \rightarrow \text{multiply, simplify then integrate.}$$

$$= x \sin x - \int \sin x \, dx$$
$$= x \sin x - (-\cos x) + C$$
$$= x \sin x + \cos x + C$$

$$2) \int x e^{3x} \, dx$$

$$u = x$$
$$\frac{du}{dx} = 1$$

$$\frac{dv}{dx} = e^{3x}$$
$$v = \frac{1}{3} e^{3x}$$

$$\int x e^{3x} \, dx = x \cdot \left(\frac{1}{3} e^{3x} \right) - \int \frac{1}{3} e^{3x} \times 1 \, dx$$

$$= \frac{x}{3} e^{3x} - \int \frac{e^{3x}}{3} \, dx$$

$$= \frac{x}{3} e^{3x} - \frac{1}{3} \times \frac{e^{3x}}{3} + C = \frac{x}{3} e^{3x} - \frac{1}{9} e^{3x} + C$$

$$= e^{3x} \left(\frac{1}{3} x - \frac{1}{9} \right) + C$$

$$3. I = \int_2^8 x \ln x \, dx$$

show that $I = 94 \ln 2 - 15$

$$\text{Let } u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\frac{dv}{dx} = x$$

$$v = \frac{x^2}{2}$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\begin{aligned} \int_2^8 x \ln x \, dx &= \left[\ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \right]_2^8 \\ &= \left[\frac{x^2}{2} \ln x - \int \frac{x}{2} dx \right]_2^8 \\ &= \left[\frac{x^2}{2} \ln x - \frac{x^2}{2} \cdot \frac{1}{2} \right]_2^8 \\ &= \left[\frac{x^2}{2} \ln x - \frac{x^2}{4} \right]_2^8 \\ &= \left[\frac{8^2}{2} \ln 8 - \frac{8^2}{4} \right] - \left[\frac{2^2}{2} \ln 2 - \frac{2^2}{4} \right] \\ &= 32 \ln 8 - 16 - 2 \ln 2 + 1 \\ &= 32 \ln 2^3 - 2 \ln 2 - 15 \\ &= 96 \ln 2 - 2 \ln 2 - 15 \\ &= 94 \ln 2 - 15 \end{aligned}$$

4. $\int \ln x \, dx = \int 1 \cdot \ln x \, dx$ when you cannot integrate $\ln x$

Let $u = \ln x$

$\frac{du}{dx} = \frac{1}{x}$

$v = x$

$= x \ln x - \int x \cdot \frac{1}{x} \, dx$

$= x \ln x - \int 1 \, dx$

$= x \ln x - x + C$

NOTE: Integration by parts twice

In some cases you will need to use integration by parts twice during the solution.

e.g. $\int x^2 \sin x \, dx$

INTEGRATION by Substitution for integrating products

Case 1 One term in the product is the derivative
(or similar to) of part of the other term

Example $\int 2x \sqrt{x^2 - 4} \, dx$

$\uparrow \quad \quad \quad \rightarrow 2x$

Method 1. Let $u =$ part of one of the terms

$$\text{Let } u = x^2 - 4$$

2. use differentiation to find a relationship
between dx and du

$$u = x^2 - 4$$

$$\frac{du}{dx} = 2x$$

$$du = 2x \, dx$$

3. Substitute u and du into the question
ie No more x

$$\begin{aligned} \int 2x \sqrt{x^2 - 4} \, dx &= \int \sqrt{x^2 - 4} \, 2x \, dx \\ &= \int \sqrt{u} \, du \end{aligned}$$

4. Integrate in terms of u $= \int u^{\frac{1}{2}} \, du$

$$= \frac{2u^{\frac{3}{2}}}{\frac{3}{2}} + c$$

5. rewrite the answer in terms of x $= \frac{2}{3} (x^2 - 4)^{\frac{3}{2}} + c$

Eg. $\int 3x \cos(x^2 + 5) dx$

$\xrightarrow{\frac{d}{dx} x^2 = 2x}$

$= \frac{3}{2} \int 2x \cos(x^2 + 5) dx$

① Let $u = x^2 + 5$

② $\frac{du}{dx} = 2x$

$du = 2x dx$

③ $= \frac{3}{2} \int \cos(x^2 + 5) 2x dx$

$= \frac{3}{2} \int \cos(u) du$

④ $= \frac{3}{2} (\sin u) + C \rightarrow \text{not } \frac{3}{2} \sin u + \frac{3}{2} C$

⑤ $= \frac{3}{2} \sin(x^2 + 5) + C$

case 2 other products

E.g $\int x \sqrt{x-2} dx$

method 1. Let $u =$ part of one of the terms

$$u = x - 2 \Rightarrow u + 2 = x \text{ (make } x \text{ the subject)}$$

2. use differentiation to find a relationship between dx and du .

$$u = x - 2$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

3. substitute (no more x 's!!)

$$\int x \sqrt{x-2} dx = \int (u+2) \sqrt{u} du$$

4. expand, simplify = $\int (u+2) u^{\frac{1}{2}} du$
then integrate

$$= \int u^{\frac{3}{2}} + 2u^{\frac{1}{2}} du$$

$$= \frac{2}{5} u^{\frac{5}{2}} + \frac{4}{3} u^{\frac{3}{2}} + c$$

5. Rewrite answer in term of x

$$= \frac{2}{5} (x-2)^{\frac{5}{2}} + \frac{4}{3} (x-2)^{\frac{3}{2}} + c$$

Case 3 Trig Substitution

$$\text{E.g } I = \int \frac{1}{\sqrt{4-x^2}} dx$$

by using the substitution $x = 2\sin u$ find I .

Method 1. Let $u =$ part of one of the terms

$$x = 2\sin u \rightarrow \text{make } u \text{ the subject} \quad \frac{x}{2} = \sin u$$

$$u = \sin^{-1}\left(\frac{x}{2}\right)$$

2. use differentiation to find a relationship between dx and du

$$\frac{dx}{du} = 2\cos u$$

$$dx = 2\cos u du$$

$$I = \int \frac{1}{\sqrt{4-x^2}} dx$$

$$3. = \int \frac{1}{\sqrt{4-4\sin^2 u}} 2\cos u du$$

$\xrightarrow{\quad (2\sin u)^2 \quad}$

$$4. = \int \frac{1}{2\sqrt{1-\sin^2 u}} 2\cos u du = \int \frac{2\cos u}{2\sqrt{\cos^2 u}} du$$

$$= \int \frac{\cos u}{\cos u} du = \int 1 du = u + c$$

$$5. = \sin^{-1}\left(\frac{x}{2}\right) + c$$

INTEGRATING FRACTIONS OF THE FORM $\int \frac{f'}{f} dx$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

$$\frac{2x}{x^2+7}$$

EX: $\int \frac{3x^2 + 2x}{x^3 + x^2} dx$

$$= \ln|x^3 + x^2| + c$$

2. $\int \frac{2x}{3x^2 + 1} dx = \frac{1}{3} \int \frac{6x}{3x^2 + 1} dx$

$$= \frac{1}{3} \ln|3x^2 + 1| + c$$

$$= \ln|\sqrt[3]{3x^2 + 1}| + c$$

$\int \frac{3x}{x^3 + 1} dx$ can't do it $\frac{1}{x} \int \frac{3x \cdot x}{x^3 + 1} dx$ X

3. $\int \frac{e^x}{e^x + 2} dx = \ln|e^x + 2| + c$

Special case

$$\int \tan x dx = - \int \frac{\sin x}{\cos x} dx$$

$$= - \ln|\cos x| + c$$

$$= \ln|(\cos x)^{-1}| + c$$

$$= \ln\left|\frac{1}{\cos x}\right| + c$$

$$= \ln|\sec x| + c$$

INTEGRATION USING PARTIAL FRACTIONS

$$\text{Let } f(x) = \frac{12 + 8x - x^2}{(2-x)(4+x^2)}$$

$$A = 3$$

$$B = 4$$

$$C = 0$$

(i) Express $f(x)$ in the form $\frac{A}{2-x} + \frac{Bx+C}{4+x^2}$

(ii) Show that $\int_0^1 f(x) dx = \ln\left(\frac{25}{2}\right)$

$$\int_0^1 \frac{12 + 8x - x^2}{(2-x)(4+x^2)} dx$$

$$\rightarrow 2 \int \frac{2x}{x^2+4} = \frac{f'}{f}$$

$$= \int_0^1 \frac{3}{2-x} + \frac{4x}{4+x^2} dx = \int_0^1 \frac{3}{-x+2} dx + \int_0^1 \frac{4x}{x^2+4} dx$$

$$= \left[\frac{3}{-1} \ln|-x+2| \right]_0^1 + \left[2 \ln|x^2+4| \right]_0^1$$

$$= \left[-3 \ln|-x+2| \right]_0^1 + \left[2 \ln|x^2+4| \right]_0^1$$

$$\rightarrow 2 \ln 2^2 = 4 \ln 2$$

$$= [(-3 \ln(1)) - (-3 \ln(2))] + [(2 \ln 5) - (2 \ln 4)]$$

$$= 0 + 3 \ln 2 + \ln 5^2 - 4 \ln 2 = \ln 25 - \ln 2$$

$$= \ln\left(\frac{25}{2}\right)$$

DIFFERENTIATION OF $\tan^{-1} x$

$$\text{if } y = \tan^{-1} x \text{ then } \frac{dy}{dx} = \frac{1}{1+x^2}$$

Proof: if $y = \tan^{-1} x$
then $\tan y = x$

$$\text{and } \frac{d}{dx}(\tan y) = \frac{d}{dx}(x)$$

$$\sec^2 y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2 y} \quad / \tan y = x$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

Example :

$$\text{If } y = \tan^{-1}(10x) \text{ Find } \frac{dy}{dx} \quad \left(\frac{d}{dx}(10x) \right)$$

$$\frac{dy}{dx} = \frac{1}{1+(10x)^2} \times 10 = \frac{10}{1+100x^2}$$

$$\text{If } y = \tan^{-1}(ax) \text{ then } \frac{dy}{dx} = \frac{a}{1+a^2x^2}$$

Find the equation of the tangent to the curve $y = \tan^{-1}(\sqrt{3}x)$ at the point where $x = 1$.

$$y = \tan^{-1}(\sqrt{3}x)$$
$$\frac{dy}{dx} = \frac{\sqrt{3}}{1 + (\sqrt{3}x)^2} = \frac{\sqrt{3}}{1 + 3x^2}$$

$$\text{At } x = 1 \rightarrow m = \frac{\sqrt{3}}{1 + 3(1)^2} = \frac{\sqrt{3}}{4}$$

$$\text{At } x = 1 \rightarrow y = \tan^{-1}(\sqrt{3}) = \frac{1}{3}\pi$$

$$y - \frac{1}{3}\pi = \frac{\sqrt{3}}{4}(x - 1)$$

$$y = \frac{\sqrt{3}}{4}x - \frac{\sqrt{3}}{4} + \frac{\pi}{3}$$

INTEGRATION of $\int \frac{1}{1+x^2} dx$

$$\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + c$$

reverse of differentiation formula

Integration formula sheet.

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \frac{5}{x^2 + 7} dx = \frac{5}{\sqrt{7}} \tan^{-1}\left(\frac{x}{\sqrt{7}}\right)$$

$$\text{Ex. } \int \frac{1}{x^2 + 9} dx = \int \frac{1}{x^2 + 3^2} dx = \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C$$

$$\begin{aligned} \int \frac{2}{5x^2 + 3} dx &= 2 \int \frac{1}{5x^2 + 3} dx = 2 \int \frac{1}{5(x^2 + \frac{3}{5})} dx \\ &= \frac{2}{5} \int \frac{1}{x^2 + \left(\sqrt{\frac{3}{5}}\right)^2} dx \\ &= \frac{2}{5} \times \sqrt{\frac{5}{3}} \tan^{-1}\left(\sqrt{\frac{5}{3}} x\right) + C \\ &\quad \swarrow \frac{1}{\frac{\sqrt{3}}{\sqrt{5}}} \end{aligned}$$

FORMULA FOR $\int \frac{1}{x^2 - a^2}$ and $\int \frac{1}{a^2 - x^2}$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right|$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right|$$

$$\text{e.g. } \int \frac{1}{x^2 - 9} dx = \frac{1}{6} \ln \left| \frac{x - 3}{x + 3} \right|$$

Choosing an appropriate method of integration

Simple cases:

$f(x)$	$\int f(x) dx$
$x^n \quad (n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x} \quad (x \neq 0)$	$\ln x $
$e^x \quad (x \in \mathbb{R})$	e^x
$\sin x \quad (x \in \mathbb{R})$	$-\cos x$
$\cos x \quad (x \in \mathbb{R})$	$\sin x$

Other integrations $\rightarrow$

Simple cases with **linear** terms:

$f(x)$	$\int f(x) dx$
$(ax+b)^n$	$\frac{(ax+b)^{n+1}}{a(n+1)}$
$\frac{1}{(ax+b)}$	$\frac{1}{a} \ln ax+b $
$e^{(ax+b)}$	$\frac{1}{a} e^{(ax+b)}$
$\sin(ax+b)$ not $\sin^2$	$-\frac{1}{a} \cos(ax+b)$
$\cos(ax+b)$	$\frac{1}{a} \sin(ax+b)$

Type of expression to be integrated	Examples	Method of integration
Product of two expressions of the form $f'(x)g[f(x)]$ Note that $f'(x)$ means $\frac{d}{dx}[f(x)]$	$\frac{2xe^{x^2}}{x^2(x^3+1)^6}$ $\frac{d}{dx} e^{x^2} = e^{x^2} \cdot 2x$	Substitution $u = f(x)$
Other products, particularly when one expression is a small positive integer power of x or a polynomial in x	xe^x $x^2 \sin x$	Integration by parts $uv - \int v \frac{du}{dx} dx$
Quotients of the form $\frac{f'(x)}{f(x)}$ or expressions which can easily be converted to this form	$\frac{x}{x^2+1}$ $\frac{\sin x}{\cos x}$	Substitution $u = f(x)$ or by inspection: $k \ln f(x) + c$, where k is known
Polynomial quotients which may be split into partial fractions	$\frac{x+1}{x(x-1)}$ $\frac{x-4}{x^2-x-2}$	Split into partial fractions and integrate term by term
Even powers of $\sin x$ or $\cos x$	$\sin^2 x$ $\cos^4 x \rightarrow (\cos^2 x)^2$ double angle formulae twice	Use the double-angle formulae to transform the expression before integrating.
Expressions in the form $\frac{1}{x^2+k^2}$	$\frac{1}{x^2+25}$ or $\frac{1}{3x^2+4}$	Use standard result: $\int \frac{1}{x^2+k^2} dx = \frac{1}{k} \tan^{-1}\left(\frac{x}{k}\right) + c$

Odd powers of $\sin x$ or $\cos x$	$\cos^3 x$	Use $\cos^2 x + \sin^2 x = 1$ and write in form $f'(x)g[f(x)]$
------------------------------------	------------	--

odd power

$$\int \cos^3 x \, dx = \int \cos^2 x \cos x \, dx$$

$$= \int (1 - \sin^2 x) \cos x \, dx$$

$$= \int \cos x \, dx - \int (\sin x)^2 \cos x \, dx$$

$$\text{Let } u = \sin x$$

$$\frac{du}{dx} = \cos x$$

$$du = \cos x \, dx$$

$$= \int \cos x \, dx - \int u^2 \, du$$

$$= \sin x - \frac{u^3}{3} + C$$

$$= \sin x - \frac{1}{3} \sin^3 x + C$$

